Valithria Dreamwalker Healing: The Effects of Using a Hunter's Tenacity Pet

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1 Introduction

There seems to be a general misunderstanding about the effects that using a hunter's tenacity pet (e.g. bear, crab, etc.) on Valithria Dreamwalker has on the healing output of holy paladins. While the exact details of this interaction have been worked out on the Hunter Discord, by user Eulogy, there are not any readily available visualizations of the interaction available, which this document aims to resolve.

For those unfamiliar with the usage of a Tenacity pet on Dreamwalker, the explanation is fortunately straightforward. Tenacity pets can train a talent called "Blood of the Rhino" that increases healing done to the pet by 40%. The general idea is for Holy Paladins to put their Beacon of Light (BoL) on Dreamwalker, but to cast their Holy Lights directly on the hunter pet so that the splash healing from Glyph of Holy Light benefits from the 40% healing buff the pet has. In a typical encounter, the effect this strategy has is simple to work out and equates to a 4% increase in total healing to the main target. However, Dreamwalker is not a typical encounter due to the presence of a healing increase effect that healers stack up. The effect, called Emerald Vigor (or Twisted Nightmares on Heroic), increases healing done by 10% per stack, and stacks up to 100 times. It is due to Emerald Vigor that the exact effects of using a Tenacity pet are not trivial on this encounter, and so rather than relying on anecdotal evidence or napkin math, let us work out the relation in an analytic form.

2 Effective Healing

We will start by writing out the effective healing done per cast of holy light on Valithria, $h_{eff,\ val}$, (ignoring any effects from Emerald Vigor):

$$h_{eff,\ val} = h_{bol} + h_{splash} \tag{1}$$

Where h_{bol} is the healing done by BoL and h_{splash} is the splash healing from the Glyph of Holy Light. Since the healing done by BoL is exactly equal to the healing per cast of holy light (BoL heal duplication is *not* affected by target healing modifiers), h_{hl}, and Glyph of Holy Light splashes 10% of the direct healing done, we can write this as:

$$h_{eff,\ val} = h_{hl} + (0.1)h_{direct} \tag{2}$$

This is where the effects of the Emerald Vigor debuff must be considered before continuing. Let us consider the general case, where $R_{ev}\,(N_{ev})$ is the healing multiplier from the debuff as a function of stack count:

$$R_{ev} = (0.1)N_{ev} + 1 \tag{3}$$

We can use this multiplier to work out the direct healing done (i.e. the healing done by holy light to the non-beacon target):

$$h_{direct} = R_{ev} m_{thm} h_{hl} \tag{4}$$

Here m_{thm} is the target healing modifier, where $m_{thm} = 1$ for a regular target and $m_{thm} = 1.4$ for a tenacity pet. Using this, the effective healing per cast of holy light is then the product of the healing multiplier R_{ev} and Equation (2):

$$\begin{aligned} h_{eff,\ val} &= R_{ev}\left(h_{hl} + (0.1)h_{direct}\right) = R_{ev}\left(h_{hl} + (0.1)m_{thm}R_{ev}h_{hl}\right) \\ &= R_{ev}h_{hl}\left(1 + (0.1)m_{thm}R_{ev}\right) \end{aligned} \tag{5}$$

It makes sense to normalize this by the healing done by Holy Light, h_{hl}, since this is just a constant term that will vary from player to player depending on gear.

$$\begin{aligned} h^{norm}_{eff,\ val} = \frac{h_{eff,\ val}}{h_{hl}} &= R_{ev}\left(1 + (0.1)m_{thm}R_{ev}\right) \\ &= R_{ev} + (0.1)m_{thm}R^2_{ev} \end{aligned} \tag{6}$$

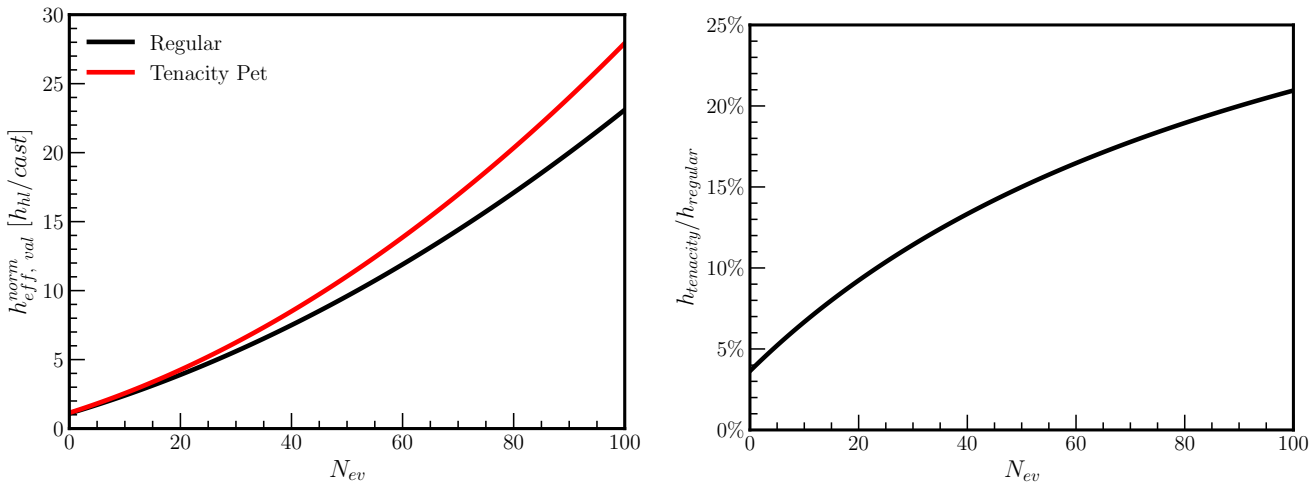
For realistic numbers of Emerald Vigor Stacks ($N_{ev} \approx 30$ at the end of an encounter), we see from Figure (1) that the healing gain from using a tenacity pet is roughly 10%. We also see that the Emerald Vigor debuff has a non-linear (quadratic) effect on the effective healing output of holy paladins. Note that, in theory, this non-linear effect favors the argument that using a Tenacity pet has a significant effect on healing output. However, as we will soon see, this is not necessarily the case due to the manner in which stacks of Emerald Vigor are accrued.

3 Simulating an Encounter

While it may seem that we can end our journey here, the picture painted in Figure (1) is perhaps a bit misleading. While a paladin may end the fight with ≈ 30 stacks of Emerald Vigor, it is important to note that they will build up these stacks in discrete intervals over the course of the fight and that the overall % increase in total healing should be expected to be less than what we see in Figure (1). In order to quantify this, we will do a bit of massaging and simplification to "simulate" Holy Paladin healing over the course of the encounter.

To do this, we will assume a few things: (1) The holy paladin will cast $C_{start} = 30$ casts of Holy light before the first portal, $C_{portal} = 12$ casts after emerging from each portal phase, $C_{end} = 30$ casts after the last portal (during Heroism), (2) the Holy Paladin will get $N_{ev} = 6$ stacks while inside each portal phase, and (3) there will be 5 portal phases. These numbers do not represent a "perfect" fight but are more representative of the performance of a "good" (these numbers were pulled from going through logs of high 90+ parsing Hpals) Holy Paladin in an average raid group. Later, we will evaluate the effects that changing these starting parameters has on the practical healing differences of using a Tenacity pet.

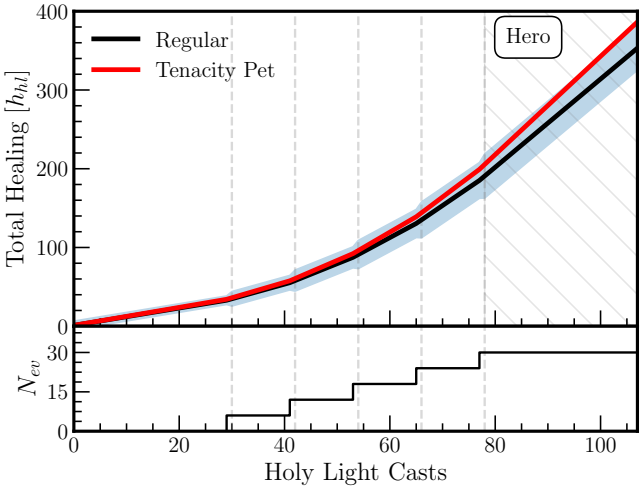
With these assumptions laid out, we can calculate the total healing done across an encounter. Since healing on Dreamwalker is sporadic, we will opt not to plot this as a function of time, but rather as a function of Holy Light Casts. We will evaluate this in bins of Emerald Vigor stacks, N_{ev}, so that we



(a) Healing per cast of HL

(b) % Gain from Tenacity Pet

Figure~1: Healing per cast of holy light for different numbers of Emerald Vigor stacks (left) and % gain from the use of a tenacity pet per cast of holy light (right). At the end of a fight, where a holy paladin will have ≈ 30 stacks, we see that there is roughly a $\approx 10\%$ gain per cast.



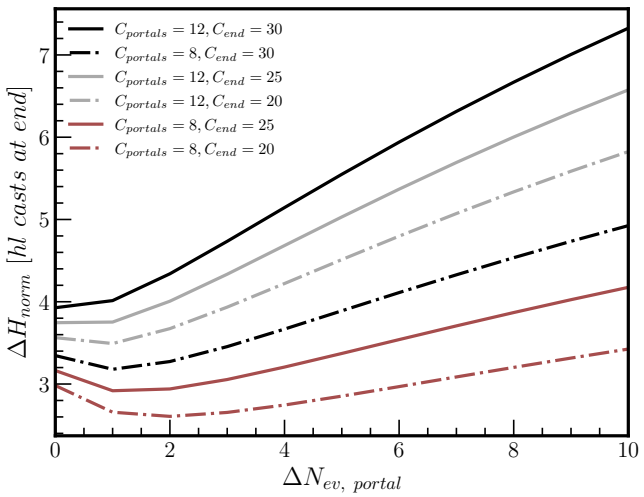
Figure~2: Total Healing in units of h_{hl} across the duration of an encounter. There were 30 cast of HL before the first portal, 12 after each portal, and another 30 casts after the last portal during Heroism. The blue band represents the effective healing on a regular target with ± 5 casts of Holy Light. Note that the total effective healing with a Tenacity pet is approximately equal to that on a regular target with 5 more casts of Holy Light.

can properly account for the effects of the debuff at each point of the fight.

We can see from Figure (2) that using a Tenacity pet equates to about +5 more casts of Holy Light (at $N_{ev} = 30$ at the end of the fight) without a regular target. While this is certainly not insignificant, note that for a Paladin in phase 3 (togc) bis with $\approx 29\%$ haste under the effects of Heroism, this equates to ≈ 6 fewer seconds, for an encounter that is ≈ 5 minutes if 5 portal phases are needed.

Another interesting avenue is to investigate how these effects change with different starting parameters (C_{portal}, C_{end}) and varying amounts of debuff stacks accrued during each portal phase, ΔN_{ev}. Figure (3) shows how many more casts of holy light (at the end of the fight) are effectively gained by using a tenacity pet as a function of ΔN_{ev} for different combinations of starting parameters.

We see in Figure (3) that a high-performing holy paladin (i.e playing well and maximizing Holy Light casts during each phase), gains a little over 7 effective casts of Holy Light, whereas a Holy Paladin performing on more of an average level, gains close to 4 effective casts of Holy Light. Thus the time saved on an encounter from using a Tenacity pet loosely ranges from $\approx 4 - 8$ seconds per Holy Paladin.



Figure~3: Effective Casts of Holy Light gained from using a Tenacity pet for different combinations of encounter parameters. From the high to low/average performance end of the spectrum, the healing gains from using a Tenacity pet loosely equates to a $\approx 4-8$ few seconds per Holy Paladin. All scenarios assume $C_{start} = 30$ casts of holy light before the first portal.

4 Conclusions

Across the course of an entire Dreamwalker encounter, it is evident that the use of a hunter's Tenacity pet effectively equates to a few more casts of Holy Light. For a paladin in Phase 3 bis, this roughly translates to an effective time saving of $\approx 4-8$ seconds per Holy Paladin. So then, is a Tenacity pet worth using? Well, in a perfect world, there's really no reason not to. Pet damage is not significant in this fight and pets can die easily if hunters do not manage them well. Additionally, while the healing gained from using a tenacity pet is not necessarily large, one could envision a scenario where a few seconds of time savings is enough to prevent a wipe and get the boss healed. However, there are some nuances that might change whether or not it is worth pursuing for your raid:

1. This analysis assumes that the Tenacity pet is stationary, close enough to the boss for HL splash, for the entire duration of the fight. Any missed Holy Light splash caused by the pet moving will negate any healing gain from the use of a Tenacity pet.

2. On Heroic, the Twisted Nightmares debuff also causes damage to the healer, causing 200 dps per stack. If the Holy Paladin heals themselves (with BoL on Valithria as usual), they can almost negate any possibility of dying.

3. If a Holy Paladin loses stacks or does not accrue an adequate number of stacks, you will lose out on much more healing than the Tenacity pet will

grant. Healer skill, in getting stacks while in the portal and maximizing Holy Light casts, is much more important.

The argument for using a Tenacity pet becomes much more convincing when there are multiple Holy Paladins to benefit from the bonuses. With 3+ Holy Paladins playing optimally, it is possible to save on order of ≈ 20 seconds off the fight. With 2 or fewer Holy Paladins, however, the healing gained/time savings are unlikely to save an encounter that would otherwise be a wipe. Even then, if the existing holy paladins are performing at a high level (i.e. getting 6+ stacks per portal phase, casting 12+ holy lights between portal phases), using a Tenacity pet is likely worthwhile. Use your own judgment and knowledge of your raid's performance and goals to decide whether using a Tenacity pet is worth it for your raid.

In the future, it may be interesting to quantify the net gains that a Holy Priest using Guardian Spirit at key points of the encounter provide, or the losses caused by missing holy light splash from the Hunter's pet moving. Upon cursory glance, Guardian Spirit should be expected to be a massive gain, as it is a flat 40% gain on all heals (there is a reason you see a Holy priest in most highly parsing Valithria clears).

"... Everything is hunter loot ..."

- Confucius